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A review of football player metrics and valuation methods: a typological framework of football player valuations

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ABSTRACT

This paper introduces a framework for football player valuations, integrating Damodaran's valuation typologies and providing a decision framework for selecting the typology.

Purpose/Rationale: It aims to refine Damodaran's valuation typologies of intrinsic, relative, real options and probabilistic methods to football valuations, highlighting current limitations of using proxies and addressing the conflation between price and value.

Design/Approach/Methodology: The paper employs a conceptual analysis of a literature review, aiming to determine valuation typologies in the literature and their limitations, aligning the literature to the established typologies of Damodaran and provides a framework for using different valuation types.

Findings: While intrinsic valuations are robust in theory, their practical application faces challenges. Relative valuations are more pragmatic and contingent on standardised metrics. Real options and probabilistic approaches present potential for future research.

Practical Implications: The framework guides researchers and practitioners when to apply different valuation typologies within sport finance, offering insights for conducting comprehensive player valuations.

Research Contribution: This paper unifies existing valuation methods into a cohesive framework, filling methodological gaps and suggesting areas for further research in football player valuations.

Originality/Value: The paper's originality lies in integrating Damodaran's approaches with sports finance, offering valuable insights for both academic research and practical applications.

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Football; player; valuation; soccer; Damodaran; real options; Corporate finance methods

1. Introduction

Football financial sustainability is currently a preeminent issue in the sport, with Martín-Magdalena et al. (2024) stating that financial issues are driven by overspending, short-term success and inflation in the transfer market. To accept transfer market inflation, we must acknowledge that football players have a fair market value with further consideration for how this fair market value can be measured.

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The approaches to valuing football players have been extensively considered in literature reviews by Franceschi et al. (2023a), Frick (2007) and Serna Rodríguez et al. (2022), with the opacity of player wages and transfer fees inhibiting the clarity of research in this field. Furthermore, the varying disciplines included in these previous reviews ranged broadly, including economics, finance, operational management, accounting, and computer science. With firm and established valuation theory within the finance discipline, it would be prudent to build upon the previous reviews by aligning the literature to the established field of valuation. Within the field of finance and valuation, there is extensive and agreed consensus on the techniques and frameworks researchers and practitioners should adopt to perform comprehensive valuations (Damodaran, 2006, 2012, 2018; Koller et al., 2010, 2020). Leveraging Damodaran's typologies, intrinsic valuations, relative valuations, real options valuations, and probabilistic valuations, this conceptual literature review seeks to construct a framework for understanding and applying football players' valuation typologies and how they align with broader finance theory. This review addresses the critical need for a structured and unified framework in football player valuation, a field lacking methodological clarity and consistency within an industry undergoing iterative and increased financial scrutiny (DCMS, 2023; Plumley et al., 2019). This review's central research question is: *"How can established valuation typologies be adapted and integrated to develop a robust framework for football player valuations?"*

In answering this question, the paper reviews and synthesises existing literature, aligning the extensive reviews by Franceschi et al. (2023a), Frick (2007), and Serna Rodríguez et al. (2022) to Damodaran's framework, unifying football player valuation approaches with corporate finance research. This review's contributions are twofold. First, it clarifies the application of finance theory to football player valuations. Second, it provides practical insights for researchers and practitioners by providing a decision framework and definitions for the valuation typology based on the available data. To consider a football player's valuation and a methodological framework to estimate this value, we must formally define what is being valued. Within finance literature, the definition of value will vary depending on the type of valuation approach used or the purpose of the valuation. Damodaran (2012) considers the role of valuation and how it differs in portfolio management, acquisition analysis and corporate finance. When we consider football players, they can be viewed as a portfolio of players (such as a team), players who are to be acquired (players to be bought or sold) and maximising squad or player value (whether strategically or performance). Within traditional financial settings, return on investment is typically the measurement of success when reflecting on adopting a particular strategy. Within football and its utility maximisation setting, performance returns relative to cost are the benchmarks of success in recruitment (Berri et al., 2023). Given the opacity of information acknowledged in European football regarding wages and transfer prices (Simmons, 2021), performance measurement provides the best universally available measure of value. We do not seek to delineate away from the cash flow-derived aspects of asset valuation, and this can be considered through the notion that teams who win more generate more cash flow (Carmichael et al., 2011). Due to their interconnectedness, sporting performance can be synonymous with cash flow generation. Therefore, our valuation definition is the performance attained relative to the acquisition cost. Players who produce a higher performance for low cost are good value, and those who produce a lower performance for higher costs are poor value, with the middle ground being fair value.

An interdisciplinary review of the literature has been performed to align the rich, unified literature in finance and valuations with the more

disparate literature in football player valuations. The remainder of the paper provides a literature review aligning existing valuation methodologies in football to Damodaran's research, this provides the basis for the development of the typological framework that follows. The final section addresses the implications of the findings, provides a decision tool for when to use a valuation typology, and presents recommendations for future research.

2. Literature review

The foundation of sports economic research in player valuations can be traced back to Rottenberg's (1956) seminal work, which introduced the concept that a player's value fluctuates between an upper and lower bound, where the lower bound is the capitalised cost of the player, albeit without a formal method for calculation, while the upper bound is the maximum price any team is willing to pay. The lower bound of cost capitalisation could include everything from wages, insurance, performance bonuses, transfer fees, coaching costs, etc or only a selection of these costs. Without a unified approach, researchers are not aligned in the determination of lower-bound capitalised costs. However, Rottenberg's framework has been consistently referenced in key studies using various approaches to lower bound cost capitalisation (Cairns et al., 1986; Carmichael & Thomas, 1993; Frick, 2011; Gerrard & Dobson, 2000; Jones, 1969; Sanderson & Siegfried, 2006; Sloane, 1971).

Scholars have also utilised seminal works from Scully (1974), and Neale (1964), and specific to football from Sloane (1969, 1971). These notions chart the course of player valuations in sports, particularly North American sports like Baseball, American Football and Ice Hockey, which have then been adapted to football (Sloane, 2006). These adaptations have led to some conflation between pricing and valuation factors, creating a disjointed literature base that departs from valuation theory in finance.

Neale's (1964) contributions laid the groundwork for distinguishing sports economics as a unique discipline, separate from traditional business economics. Neale's work delineated the boundaries where the sports economics field diverges from business economics, providing a crucial context for understanding player valuations. Through nine separate propositions, Neale demonstrated how established economic theory in business did not emerge in sport in the same way. A key example is the Louis-Schmelling paradox, where, unlike typical industries, competitors seek to eliminate rivals, sports teams however are interdependent – success depends on the existence and strength of competitors. This mutual reliance challenges traditional views of competition and market behaviour in economics. Neale (1964) and further studies (El-Hodiri & Quirk, 1971; Fort & Quirk, 1995; Kesenne, 2000; Scully, 1974; Sloane, 1971; Vrooman, 2000) have emphasised that sports economics discipline relies on the assumptions of differences between sports and mainstream economics. Some of these differences include non-profit objectives (Sloane, 1971), collaborative economic structures (Fort & Quirk, 1995) and win-maximisation above profit-maximisation (Kesenne, 2000). This paper aims to delineate further from economics towards finance theory by applying the principles of Damodaran's (2012; 2018) valuation typologies to develop a robust conceptual framework for football player valuations.

2.1 Intrinsic valuations

According to Damodaran (2018), every asset that generates cash flows has an intrinsic valuation, with the accepted method being Discount Cash Flow (DCF). DCF is the "*present value of expected cash flows over its life, discounted to reflect both the time value of money and the riskiness of the cash flows*"

(Damodaran, 2018, p. 29). Intrinsic valuations rely on the fundamentals of an asset. To develop DCF models, there needs to be accurate financial data with analysis of the asset's fundamentals. In a sporting context, asset fundamentals would be akin to the performance of the asset. Therefore, intrinsic valuations require accurate financial information and accurate measures of performance.

Gerrard (2015) summarised the literature surrounding football player valuations into two approaches: fundamental and comparative. Intrinsic and fundamental valuations are synonymous, as are comparative and relative valuations. According to Gerrard, fundamental valuation assesses a player's worth based on intrinsic factors like performance, skills and contributions to team success. Comparative valuation benchmarks a player against peers with similar attributes to gauge market price. Gerrard emphasises integrating both approaches to capture a player's true value, considering both performance and market conditions. Fundamental valuation adheres to principles of corporate finance, the assumed methodology is to employ DCF analysis to estimate the present value of an investment.

In sports valuation models however, marginal revenue product (MRP) is frequently utilised. MRP is the marginal revenue increase associated with the performance measure of a player (Depken & Wilson, 2004). With difficulties in determining DCF valuations for players based upon the valuation method, MRP becomes the de facto intrinsic valuation method in sport (Gerrard, 2015), with an abundance of research in sport using MRP (Bradbury, 2013; Dobson & Gerrard, 1997, 1999; Gerrard & Dobson, 2000; Krautmann, 2013; Scott et al., 1985; Scully, 1974). A player's value is measured against their individual performance within the team and how their presence increases the expected revenue returns (attendance, sponsorship, merchandise, etc.). Scully (1974) pioneered the approach of examining individual player performances and salaries in Major League Baseball (MLB). Scully determined that although it was difficult to correlate a player's performance and club revenue directly, estimating player performance and team winning was possible, then determining how team winning drives club revenue. This approach proved fruitful in North America with research conducted in sports such as baseball (Fort et al., 2019; Solow & Krautmann, 2020), American football (NFL) (Leeds & Kowalewski, 2001) and with applications in ice hockey (NHL) (Richardson, 2000) and basketball (NBA) (Simmons & Berri, 2011).

A mixture of comparative and fundamental approaches has also been observed, as Krautmann (2018) demonstrated with an MRP approach combined with a "wins above replacement" (WAR) approach. WAR is a metric that considers team performance should another player of differing value replace a particular player. A similar study was also conducted where performance was assessed against "wins above benchmark" (WAB), which is like WAR but compares team performance based on the probability of winning and match outcomes (Gerrard, 2007). An increasingly used approach is Plus-Minus rating which in principle is similar to WAR, however Plus-Minus is more applicable to fluid team sports like NHL and NBA. Plus-Minus focuses on team scoring differences, as opposed to WAR, which aims to encompass more data and isolate their contribution to the team (Fearnhead & Taylor, 2011; Kharrat et al., 2020; Macdonald, 2012). Applications of Plus-Minus in football have been presented by Kharrat et al. (2020), who also provide an overview of the history and evolution of the metric, and by McHale and Holmes (2023), specifically to the estimation of transfer fees. In the past decade, the MRP approach still sparks debate among academics, with Krautmann (2013) and Bradbury (2013) arguing it's use in a free-market approach. Krautmann (2013) advocates a free-agent market approach in determining fair player wages and argues that salaries should reflect an athlete's

contribution to team revenues without external restriction, such as regulations in the labour market, claiming MRP is an incompatible tool for understanding player wages. Bradbury (2013) is more critical of the free-agent market approach, highlighting that the sports labour market is not truly free and Scully's (1974) approach still has merit. Notwithstanding the market structure, what is clear from the research conducted on North American sports is that using MRP requires a unified measure of performance and accurate wage data.

The application to football is less robust compared to research on North American sports, primarily due to the limited availability of salary data and the performance metrics employed (Simmons, 2021). Due to this lack of publishable wage and transfer data, a number of researchers use proxies for salary, transfer fees and other monetary-related data. Many scholars rely on publications like *Kicker* magazine (Deutscher & Büschemann, 2014; Frick et al., 2010), more recently but less researched *Capology* (Berri et al., 2023) or *TransferMarkt.de* (Bryson et al., 2013, 2014; Campa, 2022; Depken & Globan, 2021; Franck & Nüesch, 2012; Frick, 2011; Rubio Martín et al., 2022). Prockl and Frick (2018) determined that the estimates of wages and fees on *TransferMarkt.de* are a good proxy, although this is contested by Coates and Parshakov (2022), Thrane (2019) and McHale and Holmes (2023). The doubts discussed in the inaccuracies of this data source are subjective bias and selection bias (Coates & Parshakov, 2022; McHale & Holmes, 2023). An issue not discussed are issues revolving around informational cascades (Bikhchandani et al., 1992) and the effects of the anchoring and adjustment heuristics (Tversky & Kahneman, 1974). The anchoring heuristic is a cognitive bias where people rely heavily on the first piece of information they encounter (the "anchor") when making decisions, causing their estimates or judgments to be influenced by that initial reference point. The adjustment heuristic is a cognitive bias where people make decisions by starting from an initial value and then making incremental adjustments, often insufficiently, leading their final judgment to be overly influenced by the starting point. Both areas warrant some scholarly review of these effects on *TransferMarkt.de* data. Researchers believe that *TransferMarkt.de* benefits from the "Wisdom of Crowds" established by Francis Galton (Galton, 1907). This was considered in a paper by Coates and Parshakov (2022) where they found the transfer fee estimates on*TransferMarkt.de* to be biased. Our main contention against the assumption of the "wisdom of crowds" effect is that *TransferMarkt.de* estimates for player valuations should include diversity of opinion, independence, decentralisation, aggregation and trust (Surowiecki, 2004). Independence cannot be observed in the process as the contributors know all prior valuations. By knowing prior estimates, new valuations will suffer from the anchoring heuristic (Tversky & Kahneman, 1974). This small, but significant violation of the "Wisdom of Crowds" would state that the website data cannot benefit from this phenomenon. Furthermore, research has shown there is accuracy in *TransferMarkt.de* valuation estimates, but there are not enough robust processes in the methods to exclude the potential for information cascades in the empirical research conducted. Of the publishable financial data in some countries (United States and Italy, section to follow), it is not clear if the data has influenced the estimates uploaded by experts, skewing the accuracy of all estimates. Research suggests that some clubs use *TransferMarkt.de* information in their negotiations (Ackerman & Follert, 2019; Coates & Parshakov, 2022), potentially creating the same effect in the opposite direction. It is not clear in the research how much effect is created by the interchangeable use of *TranferMarkt.de* and clubs. There is a tendency in the research to reverse engineer how estimates are made on *TransferMarkt.de* to ascertain the accuracy of the values depicted on the

website (Müller et al., 2017; Prockl & Frick, 2018; Richau et al., 2019; Serna Rodríguez et al., 2019; Singh & Lamba, 2019). We note from Herm et al. (2014) that playing data is used in the decision-making algorithm of the website. For researchers to then use playing data as a basis for determining valuations' accuracy does not deliver new information; it merely confirms that certain playing data plays a role in the estimation of valuations on the website. The only tangible insights are the specific data that is being used to drive the valuations on the website. As noted in Coates and Parshakov (2022), *TranferMarkt.de* states their method "*is not scientific*". Even though scholars have highlighted their doubts, recent research has used this proxy for salary data, performance data or both (Metelski, 2021; Poli et al., 2022; Prockl & Frick, 2018; Richau et al., 2019).

Two countries with differing systems of information capture and dissemination are the United States and Italy. Salary data is required under sporting regulations in the United States Major League Soccer (MLS). However, little research has been conducted on football compared to other North American sports. Medcalfe and Smith (2018) conducted their research not using wage or transfer proxies, but actual wage data published via the MLS. They used the MRP method alongside real wage data to build their findings, a departure from the use of proxies. Italy is another country where wage data is published due to a legal requirement in the country. Bryson et al. (2014), Carrieri et al. (2018) and Fumarco and Rossi (2018) all present work on wages and performance. Even though the data is available, there are issues with the reliability of the data and the application of the findings outside of Italy. In 2023, Juventus, Italy's largest and most successful team, was prosecuted for false accounting (Piovaccari, 2023). The conviction is in relation to alleged falsification of how they calculated capital gains on player transfers, by deflating purchases and inflating sales to circumvent football regulations. This does create some opacity for any research conducted using Italian club accounts.

The second element for intrinsic valuations is to consider a reliable measure of performance in football. A significant amount of research has been conducted on the projection of player salaries and transfer fees from playing data (Bell et al., 2022; Bryson et al., 2013, 2014; Carmichael et al., 1999; Carmichael & Thomas, 1993; Coates et al., 2020; Di Domizio et al., 2024; Franck & Nüesch, 2012; Frick, 2011; Lucifora & Simmons, 2003; McHale & Holmes, 2023; Metelski, 2021; Pappalardo et al., 2019; Poli et al., 2022; Szymanski, 2000, 2001; Yang et al., 2024; Yiğit et al., 2020). There is not a consensus on how to determine performance indicators, talent or ability in football, rendering the use of MRP difficult. Authors have tended to use proxies for performance measures including seasonal publications (Carmichael et al., 1999), club data (Lucifora & Simmons, 2003), websites (Bryson et al., 2013, 2014; Franck & Nüesch, 2012; Frick, 2011), computer games (Butler & Coates, 2022; Coates & Parshakov, 2022; Yiğit et al., 2020), football-specific data companies (Pappalardo et al., 2019; Tunaru et al., 2005), and as previously discussed, the use of WAR, WAB and Plus-Minus (Fearnhead & Taylor, 2011; Gerrard, 2007; Kharrat et al., 2020; Macdonald, 2012; McHale & Holmes, 2023; Saebø & Hvattum, 2019). This variation of methods highlights the issue of uniformity in performance measures.

Although most scholars have utilised MRP as an intrinsic valuation method, Leifheit and Follert (2021) present a DCF method from the corporate finance school of thought (Brealey et al., 2020). This is a rare example of where intrinsic valuation approaches are applied directly to the valuation of players. The authors assume that the cash flows from the players on contract can be accurately assessed and, therefore, discounted to the present value. Accordingly, a buyer of the asset would not pay more than the present value of the

cash flows. For a robust DCF model, there are four assumptions that need to be included: (1) accurate free cash flows; (2) risk measure; (3) growth rates; and (4) a terminal value (Damodaran, 2018). With the reduced level of accessibility to financial data, it is difficult to assess free cash flows within a football club and difficult to apportion this to a player's performance. A risk measure is highly subjective in that what exactly is the risk being measured? Is this the risk of underperformance, the risk of the player not being able to play (through injury, etc) or the volatility of performance? Risk is this context requires a much more robust definition. Growth rates would require an accurate measure of future performance and without a unified consensus around a performance measure, it is difficult to establish any growth rate. Finally, due to the Bosman Ruling, the terminal value of a player at the end of their contract is zero[1] (Antonioni & Cubbin, 2000; Simmons, 1997). We believe applications of the Plus-Minus system of Kharrat et al. (2020), McHale and Holmes (2023) and Saebø and Hvattum (2019) to DCF could provide an answer to the expected cash flow variations in DCF modelling, however, it is still a long leap from this Plus-Minus score to the effects on cash flow to the club. Another alternative to a DCF approach was explored by Follert and Gleißner (2024), who developed a present value model using cash flow variations as the measure of risk, a principle commonly seen in corporate finance. This model considers CAPM, risk-free rates, risk measurement as standard deviation, and the present value of future cash flows blended with subjective value theory (Franceschi et al., 2023b). This modelling approach is innovative in player valuations and is a stark example of the potential to align the player valuation field with the established finance field. Although only tested on fictitious data, it provides an example for practitioners privy to accurate financial data, a method aligned to the established finance field. In concluding this section, an important point to raise is that "*[a]ssets that do not produce cash flows cannot be valued using discounted cash flow models*" (Damodaran, 2012, p. 775). We argue that what must be considered within a broader discussion is the applicability of football players being classed as cash flow generating assets in the traditional sense. Although attempts to measure revenue from players' performances are indirectly noted, we suggest that the methodologies and assumptions in a football context appear tenuous. We therefore argue that applying DCF techniques to football player valuations is not tenable with the current level of financial data. Although effective in many other sports, MRP is still to be applied with conviction in football. This is due to estimating and calculating the revenue increase from a player's performance, the accurate measurement of a player's performance within team performance and the availability of reliable and accurate wage and transfer data. These points affirm Scully's (1974) concerns and support our position that restricted accurate financial information or consensus for performance determinants demonstrates intrinsic valuation approaches' limitations.

2.2 Relative valuations

Damodaran (2012) suggests that the best approach to valuing non-cash generative assets is through a comparative approach, also known as a relative valuation. Damodaran provides a simple theoretical process for how you would conduct a relative valuation. First, you need to collect a group of comparable assets. Second, estimate measures of standardised variables for this group. Third, controlling for differences between assets in the group and the asset being valued is necessary to ensure the measures are considered reasonable. This

[1]The Bosman Ruling determined a player was not bound to any club when their contract expired, creating free agency in football.

is a simple theoretical process, but it is not as simple as suggested in practice. Damodaran (2018) states that finding comparable assets is difficult. Markets for these assets are not usually liquid or public, with many private transactions. It is also unclear how one controls for the differences across the assets and prices for these assets are often related to the differences or scarcity of the asset.

Forming a standardised approach is described as converting prices into multiples of some common variable (Damodaran, 2012). In the absence of price data, we argue that the standardisation of common variables across comparable players could be applied. For example, if we assume strikers are a group of comparable assets, the standardised approach would be to determine the comparable variables across this group so that all strikers are compared under the same constraints. This would support Damodaran's process's first and second points by constructing groups of comparable assets and including the same common variables across this group. This approach was demonstrated by Berri et al. (2023) in their research around the goalkeeping group and the salary determinants of their performance variables. Controlling for differences in the group so that the measure is reasonable would be the final step and one that has been researched well in the literature (Bryson et al., 2013; Carmichael et al., 1999; Carmichael & Thomas, 1993; Dobson & Gerrard, 1997, 1999; Gerrard & Dobson, 2000; Szymanski, 2000; Wicker et al., 2013). The most common variables used in player valuation are age, nationality, playing level, experience and performance, with control for differences often deriving from one of these variables.

At a superficial level, a relative valuation approach may be more suitable for football player valuations than intrinsic valuations, as we are unable to focus on player derived cash flows or proxies for this. However, we argue that the flaw in the relative valuation approach is that without clear knowledge of prices paid for other players, it is questionable whether a valuation can be determined when prices for comparable players are unknown. We will revert to this point at the end of this section.

Other valuation models that have sought to utilise a relative valuation approach have used various proxies. Campa (2022) researched the difference between transfer fees and the estimated inherent value of players. Although not formally described in the paper, inherent value is inferred as the quality of the player. A two-part approach is used to assess asset quality: first, by considering the price paid to acquire the asset; second, by evaluating the difference between the purchase price and the asset's inherent value. In this paper, Campa used *TransferMarkt.de* as the metric for both the value and the transfer fee paid, with the value being what *TransferMarkt.de* ascribe to each player through their algorithm (discussed previously). Although the author's approach did consider an intrinsic approach, the ultimate aim for value determination was to compare the proxy valuation (TransferMarkt.de) with the author's interpretation of inherent value. Noting the limitations of this data source identified earlier, we argue the evidence suggests a discrepancy between the price paid by clubs in the forms of transfer fees and the inherent value of the player, as determined by their *TransferMarkt.de* data. This position supports the relative valuation approach outlined by Damodaran (2012) but also provides evidence that there are differences between the valuations and pricing of players. A point we will revert to later.

One area of comparative valuation often used to measure risk is the certainty or consistency of cashflows from various investments (Damodaran, 2012). As noted from our definition of value, cash flow measurement from players is difficult. However, performance levels are correlative with cash flow. Therefore we argue that performance consistency can be considered as a proxy for cash flow

consistency. Deutscher and Büschemann (2014) examined performance consistency and player wages. In their paper, a well-crafted literature review surveyed the factors associated with performance linked to pay, concluding that players who perform inconsistently were granted a salary premium. Their concluding comments highlight their findings contradict Bodvarsson and Brastow (1998) and suggest the reason could be associated with the data they used. For example, with football being a low scoring game, a player may score one goal every two games on average, however these goals may come in batches (i.e. scores 10 goals in 10 games, then 0 goals in 10 games etc). At the end of a regular 38 game Premier League season, a return of 19 goals would be deemed exceptionally good, placing them in the top 10 goal scorers, every year since the Premier League's inception (Premier League, 2023). The same would be true of the playing data metric of assists. Ryan Giggs, the Manchester United stalwart who completed 632 Premier League games for the club, has the highest number of assists of any Premier League player, 162 in total. This gives a metric of 0.26 assists per game. This is a very low number and provides scope for the volatility of performances attributed to why a wage premium was noted in their research. Following the same logic for the goal scoring metric, an assist every other game would provide a ratio of 0.5 assists per game, nearly double that of Ryan Giggs. Assuming the same batch approach (10 assists in 10 games, then 0 in 10 games), this performance volatility would still generate superior performance compared to other players. Deutscher and Büschemann also considered the points distribution in football as a potential factor in the results due to wins scoring much higher than draws and losses (3, 1 and 0 points). In a game where a player could score four goals, the team may also concede four and the resultant points scored for the match would be one. Alternatively, a player may create one assist in a match that finishes 1–0, and that assist scores the team three points. The superior actions in one match for a player do not equate to the points of a mediocre action of another player in another match. We argue that match results are not a determinant of individual performance but group performance and highlight the difficulties in quantifying performance or producing a standardised measure.

Performance is a tool for measuring value and therefore, a measure of players' wages. Szymanski and Smith (1997) unequivocally determined that pay and performance are inextricably linked. Szymanski and Smith (1997) discuss how efficient a team was in their spending as a measure of quality for the resourcefulness of the club's recruitment. Teams who paid more money for better talent and accrued more points used a correct recruitment strategy, and those who paid more money for better talent and accrued fewer points used an incorrect strategy. This can lead to the notion that merely paying more wages is the key to higher performance, which is not the point Szymanski and Smith determined in their paper. To secure superior current and future performance of a player, there is a requirement to pay higher wages. The critical part of the process is the accuracy of determining current and future performance. The wages paid are the outcome of performance prediction; only after the fact can both overpayment and underpayment of wages or over-prediction or under-prediction of performance be considered. The net difference of this is the proxy for team or player quality of past performances. Furthermore, with real-time wage data not available, it is not possible to determine the current market price for wages relative to current performance when determining contracts. We argue that as the information is *ex post*, it does not help determine current or future valuations from a relative valuation perspective.

Performance can be measured in different ways, with a view of the past performances previously discussed, but also potential future

performances. Gerrard (2001) developed a model to measure player and team quality in professional team sports. His paper provides a useful overview of the economic applications of player and team valuations, citing the previous work by Dawson et al. (2000). Gerrard suggests that the production function of a team is dependent on the production function of the players, plus coaching input and unspecified factors. Gerrard names the approach Player Quality Index (PQI) and it is described as a set of observable characteristics, such as age, experience and previous performance levels, that are all measured prior to the next performance. The approach allows for *ex ante* valuation and can be used as a metric for individual player quality. The paper provides a simple table ranking the players assessed from 1 to 20. The highest scoring player listed in the paper, Andy Cole, had a PQI of 461.89 and 20th placed player Dion Dublin had a PQI of 313.94, with players ascending in value to the highest valued player. Quality, however, is not the same as performance. Gerrard (2015) presented a 3×2 matrix as an attempt to capture performance through objective data with classifications of positive or negative contributions to team performance. STARS (Statistical Trend Analysis Rating System) allows quantitative methods and some qualitative input to generate a rating for a player in invasion sports such as football. The system was trialled and calibrated on 760 Premier League matches between 2004 and 2006. We suggest that this system appears to be an improvement in subjective bias that can be introduced in a performance grading environment that utilises sufficiently agreeable quantitative data to support the subjectivity.

Performance measurement can appear intuitive, however as noted by Damodaran (2018), comparative valuations can be difficult when deciding what factors are indeed comparable and measurable. Müller et al. (2017) demonstrate there are multiple player-specific factors that determine a valuation for the player. Player performance, player characteristics and player popularity were used as variables, combined with market values from *TransferMarkt.de*. The resultant model attempted to provide a framework to develop market values for players. In following the work of He et al. (2015), there was a tenuous leap to consider these valuations as prices, further evidencing the conflation between price and value. Price factors involve different variables than those to derive valuations. Price factors would be contract length, financial positions of both clubs and, more recently, regulatory constraints to meet thresholds. We agree it is correct to state that valuations form the basis of a price determination strategy, however, we would suggest that valuations are determined from prices would not be a correct relative valuation approach. We argue that value is driven by on field performance and player characteristics like age and experience only and is not influenced by the price factors noted above. For example, we suggest that research focusing on valuation should not include contract length as contract length is a determinant of price, not value. There are various claims that the Bosman Ruling has influenced player valuations (Antonioni & Cubbin, 2000), but we argue that this is axiomatically wrong. Contract lengths and factors related to the supply and demand of players are research that focuses on price determination (Damodaran, 2018). We argue that there is a conflation of information on player, buying and selling club characteristics on how they can influence a valuation. Some club characteristics will drive up valuations; for example, a winning team will undoubtedly influence player valuations, but more so because the player would be deemed a winner in their own right. Similarly, we suggest that research that uses the financial characteristics of a club as a driver of players' value is incorrect in its approach. The financial characteristics of clubs will affect the prices paid and the negotiations but do not influence the player's value.

There has been research that has considered the conflation in price and value with varying degrees of depth and examples (Detzen & Löhlein, 2023; Follert & Gleißner, 2024; Franceschi et al., 2023b; Quansah et al., 2021); however, researchers are not yet entirely distinguishing between the two concepts. By way of an example, we can highlight the issue of using prices paid and club characteristics to ascertain a player's value. Player X is being considered for a transfer from Town FC to City FC. City FC have just received a significant investment in cash from sold equity and can now build a better squad with this new capital. Town FC are negotiating to sell their superstar player, Player Y, with Player X, the second-best player at Town FC. City FC have placed a bid of £10 million for Player X, their price based upon their perceived value for the player. This example removes the negotiation and external influences on their offer. Town FC are mindful of Player Y's negotiation with another team and is awaiting the conclusion of the negotiation before looking at City FC's bid. The four scenarios that Town FC are considering are, (1) selling Player Y for £20 million and using this money to strengthen their squad, therefore keeping Player X as part of this process, (2) not selling Player Y but selling Player X for £10 million and using that money to strengthen their squad, (3) sell neither Player X or Player Y as they are both under contract and try for a better league performance next season, (4) sell both Player X and Player Y and use the money to rebuild the squad. City FC are keen to have their squad in place for pre-season and insists Town FC decide on their offer. Town FC has several weeks of the transfer window remaining and is prepared to wait. The only move City FC has to build their squad early is to pay more money or look for another player.

Player X has not played since City FC prepared their £10 million offer based on their valuation of the player. Due to Town FC having different scenarios they are yet to explore; they have not come to a decision on City FC's offer. At this stage there is no confirmation if this amount of money is below, at or above the player's value from Town FC's perspective. As Town FC have the most information on the player, they are in the best position to ascertain the value of the player and without their response, there is no way of knowing how accurate the £10 million valuation of City FC is. Should City FC decide to increase their offer and Town FC accept, recently published models (Campa, 2022; Poli et al., 2022) suggest the price paid is proportionate to the player's value and should be used within models that aim to determine the values of other players. We suggest that this disparity is why value and price are conflated and demonstrates that the characteristics of buying and selling clubs are more relevant in price determination but not as relevant in valuations.

Relative valuations provide a robust basis for use in player valuations. Ensuring the three rules laid out by Damodaran (2018) are met, this approach provides a pragmatic basis to determine value. What is vitally important is that prices are not deemed comparative with values and conversely, associated drivers of value and prices are not assumed interchangeable. As with intrinsic valuations, much of the research in this field is subject to selection bias. Players who have previously been transferred and those who were not transferred should not share the same probability of being transferred (Carmichael et al., 1999; Monteiro et al., 2023). The factors considered in the transferred players and non-transferred players may be syllogistic across the two groups, but this assumption is not axiomatic. Furthermore, relative valuations in traditional valuation theory do not concern themselves with factors determining the price (e.g. supply and demand). Future relative valuation research should not constrain itself to only the pool of limited players who transferred but also the pool of players who can be transferred. This will strengthen the relative valuation research

through its alignment to mainstream finance but also will provide the ability to consider a market of available players and allow for a fair market valuation, extending the research in broader finance theory, as shown in Follert and Gleißner (2024).

2.3 Real options valuations

Real options valuations are described by Damodaran (2018) as a valuation approach that considers the upside of risk. Damodaran (2024) noted that the most difficult intangible assets to value are those that have the potential to create cash flows in the future but do not right now. We argue that a football player has the potential to generate future cash flow, whether by transfer, commercial revenue or individual sponsorship. With the potential for future cash flow from these sources, this displays option characteristics (Damodaran, 2018), which could apply to player valuations. For example, with Relative Valuation and Intrinsic Valuation, the risk is like a depressant on the valuation potential; the higher the risk, the lower the potential valuation. With real options, risk serves as a measure of the opportunity and provides the valuer with a view of the potential upside given the risk and the costs associated with taking that risk. Real options can be viewed as decision trees where the probability of a given outcome can determine the expected valuation of the decision. If the expected value is negative, normal practice would see that the investment option is rejected. In practice, decision trees are more complex, with a range of possibilities and various payoffs.

In academia, options valuation research gained momentum after the publication of the Black–Scholes model, eventually providing a Nobel Prize[2] for the research (Black & Scholes, 1973). Stated simply, the Black–Scholes model is a Partial-Differential Equation that assumes the price of the option as a function of the stock price and time, factoring the stock's volatility and risk-free rate. Under a market of zero costs to acquire the assets, the model has demonstrated that one can perfectly hedge the market and eliminate risk from investing. Therefore, the model provides an answer as to the correct price for the option. Although the model has its critics (Haug & Taleb, 2011), it is still widely cited in empirical research as a method for valuing options (Weatherall, 2018). Real options were established by Myers (1977), who considered that a firm can be viewed as a construct of real assets and real options. Real assets are tangible assets such as property, plant and machinery, etc., or intangible assets such as player contracts, goodwill, etc., but real options were deemed opportunities to buy real assets on terms favourable to the company.

In the context of player valuations, there are a surprisingly small number of applications of options valuation techniques. Tunaru et al. (2005) were the first to use an options pricing framework from the finance literature (Black & Scholes, 1973; Merton, 1973; Samuelson, 1965) to determine transfer values and did not build on valuation-related research from Rottenberg (1956) or Sloane (1969, 1971). Using the Opta Index[3], the model was tested on one Premier League player. The model used buying club characteristics, player characteristics (in the form of the OPTA Index score), Markov Processes (Brownian Motion) and Poisson's Law to correct for gaps in the playing data. The model showed an evolving player valuation. This paper was developed as a framework for conducting a valuation process.

Tunaru and Viney (2010) applied this framework to assess the valuation of the same player used by Tunaru et al. (2005), but also to assess if the player would have a different value with a different club. Thierry Henry was valued at

[2]As the Nobel prize is not awarded posthumously, Fischer Black was not awarded the prize but was given credit for his role.
[3]Opta Index is a composite measure of player performance, calculated by a private company, now a subsidiary of Stats Perform.

their parent club, Arsenal, and then was compared to valuations at league rival clubs Manchester United and Aston Villa. This paper demonstrated that players would have different values at different clubs based upon factors associated with the player's performance levels and the performance of the valuing club. Later research from Kanyinda et al. (2012) followed a similar approach to Tunaru and Viney (2010) and Tunaru et al. (2005), however, they included the notion of an average player (typical) and a superstar player (atypical) to determine the differences between option pricing for these classifications of players. Similarly, Coluccia et al. (2018) utilised Tunaru et al. (2005) approach in their examination of one Italian Serie A goalkeeper, anonymised to "Alfa," playing for "Beta FC." As with Kanyinda et al. (2012), Tunaru and Viney (2010) and Tunaru et al. (2005), the authors also found credence in the methodological approach; however, like the previous research, a remarkably small sample size of one was used.

The models showed that a player's value at a specific moment in time can be predicted and is only limited by the information used in the model and the time parameter of the valuation. We argue, however, that it is unclear how accurate the predictions are due to the data and sample sizes. We suggest that the key limitation is knowing what variables influence player valuations and the time parameter for the option. Using a composite variable by a data supplier or the difference between the typical and atypical has uses for providing a framework for how valuation can be determined, but the accuracy of this composite variable was not determined. With a sample size of four players used across the three papers, we suggest it would be difficult to claim that this method has greater accuracy than others. The widespread use of options in finance and the significant empirical research around this form of modelling make this approach interesting. We believe that if it is possible to develop consensus on the value driving variables, combined with a uniform application across all players, these models do have potential to move beyond its current theoretical proposition.

2.4 Probabilistic valuations

With probabilistic valuations, the literature on football player valuations is more disparate than the previous techniques. Damodaran (2018) noted that the key to these approaches is the probability of future cash flows from the investment. The cash flow's likelihood and size are the key drivers in this valuation process. As already discussed, players are non-cash flow generative assets. This does not mean probabilistic techniques cannot be used in football player valuations but we revert to our definition of value as performance attained relative to acquisition costs. Specifically, probabilistic valuations should consider the future likelihood of performance levels or the probable acquisition cost of players. There is an overlap with the Intrinsic, Relative and Options valuation methods. However, the distinguishing difference is this section considers the approaches used that can provide probabilistic valuations via the methods used, whether a method can predict future performance or future acquisition cost or is capable of capturing both future performance and acquisition cost.

Although Damodaran does not explicitly state this, we suggest that Machine Learning (ML) techniques would fit this valuation approach, as ML relies heavily on probability (Richau et al., 2019; Singh & Lamba, 2019). Game Theory (the study of strategic decision-making where individuals or groups make choices that potentially impact the outcomes for all participants involved) applications are the first wave of probabilistic valuations seen in football. Pioneered in football by Carmichael and Thomas (1993), the valuation model relies on the transfer fee being a Nash Bargaining Solution (NBS) with the negotiation framework by Nash (1950, 1953) employed. Although Game

Theory and Probability Theory are distinctly different, a probabilistic element is present through the requirement for rationality under Game Theory (Kadane & Larkey, 1982). This rationality allows modelling for predictability and why Carmichael and Thomas (1993) can be credited with the first application of probabilistic valuations in football.

Pappalardo et al. (2019) used a machine learning approach to develop a new metric of player performance called *PlayeRank*. The authors used the *Wyscout* system, which is a database of player performance metrics widely used by professionals but seldom used in research. The research was split into three phases: rating, ranking and learning. The research covered 64 seasons, 19,619 matches, and 31 million events across 21,361 players; the sample size is extensive. The results presented some cursory accurate results with players whom you expect to be ranked highest in their role, being top (Lionel Messi, Cristiano Ronaldo, Neymar, Mohamed Salah, etc.). The *PlayeRank* system would provide a good basis for a performance measure in football, and potentially, a measure of value. However, we suggest that even with the sophisticated techniques used by Pappalardo et al. (2019), these techniques still struggle to deal with football's low-scoring nature and the disproportionately large impact a solitary goal can have. Although well established in research that strikers earn most, then midfielders, then defenders followed by goalkeepers (Medcalfe & Smith, 2018), when trying to rank players across positions, goals distort the ability to rank players. There are several other streams of machine learning and performance prediction in a football context, but much of this is geared towards team performance prediction, with the main utilisation being for gambling practices (Constantinou & Fenton, 2017; Rodrigues & Pinto, 2022).

Aydemir et al. (2022) sought to determine transfer valuations from an ML approach. Notwithstanding the conflation between a transfer valuation and transfer price, the paper did present differing views in this field. It is noted by Aydemir et al. (2022) that buying club characteristics is not feasible as the information is unavailable; this suggests that the data required to be used in the model would need to be current data. Annual accounts in the United Kingdom are published around 6–9 months after the year's end, and by the time the modelling takes place, the data no longer reflects the club's financial position. According to Aydemir et al. (2022), a player's value is a cumulative sum of the salary the player has received across their career, plus the cumulative transfer fees paid.

The Aydemir et al. (2022) paper does present a stark example of the levels of conflation between price and value. The methodology for this research took information from *Wyscout* for performance data, *TransferMarkt.de* for transfer data and *Google Trends* for player popularity data, to build a new database. An interesting feature is that even though machine learning techniques are used, there is still the propensity to transform the data to ensure it is normally distributed. Even though non-parametric techniques are used frequently in machine learning, data transformation and normalisation to correct for the positive (right) skew is unusual in this context. This skew is common in the research around player valuations and is demonstrated by Müller et al. (2017) and Poli et al. (2022), who adopted a statistical approach, and by McHale and Holmes (2023), who used an ML approach. With modern computing power there is little reason or understanding for why researchers do not adopt non-parametric methods to build upon the literature in this field, especially with the current methodological approaches already well established utilising logarithmic transformations and parametric methods.

McHale and Holmes (2023) specifically looked at transfer fees and the determinants of this using an advanced performance metric combined with machine learning. Using a

two-step approach, McHale and Holmes sought to conclude performance metrics for players, then modelled transfer prices that a club could be willing to pay. McHale and Holmes make two comments that provide insight into the different approaches used in price determination and valuation: (1) is the seeking of "value for money" transfers, and (2) is the assumed market efficiency. We make two points concerning these comments. First, although the authors state the method is to model transfer fees, they also strayed into the realm of stating potential values, and as we have noted, these are fundamentally different concepts. A similar two-step approach is required to establish if a player could be considered "value for money," the first is a valuation, and the second is a price. Without known prices, it is difficult to determine "value for money." Second, assuming market efficiency in football would not be a tenable position given the opacity within the transfer market, and although Fama (1970) provided tiers of market efficiency, it would be very tenuous to suggest the transfer market in football is in any way market efficient due to the secrecy of transactions. Playing performance is indeed a fundamental driver of value, and the Plus-Minus system used in this paper to determine playing performance is empirically tested, robust, and delivers pragmatic results. A comparative approach using a simple ranking system to determine players based on this system was also noted and would be classified as a useful relative valuation approach. When considering the transfer fees paid within the modelling process, it allows for a time-series and backdated view of establishing what fee was paid at a given moment in time based upon a player's ranking measure. We suggest the issue with this approach for valuations is that many factors determine the prices paid for players that are not necessarily reflective of the player. As noted in the Town FC and City FC example, the factors influencing the price paid move beyond the player's ability on the field and beyond that captured in a Plus-Minus system. Unless the model can control for specific reasons why a fee was paid (i.e. contract length, urgency, capital balance, decision makers), we suggest that prices do not provide a rational basis for valuations.

3. A typological framework of football player valuation

To advance the field of football player valuations, we have developed a decision framework to guide researchers in identifying the type of valuation method they should consider using based on the type of data the researcher holds or can access. This framework (see Figure 1), we believe, will allow researchers to align their work to broader finance theory and the valuation techniques developed by Damodaran (2012, 2018, 2024). Furthermore, we have provided expanded definitions, potential methods and sources of previous literature that have used that method.

Following our position that player value derives from on-field performance and player characteristics like age and experience, this decision framework allows researchers to efficiently align their valuation typology based on the data they hold. The decision processes rely on two data types: measuring cash flow or measuring playing performance. In the absence of an effective measure of player derived cash flow, we argue the most effective data in measuring value is derived from the performance data, age of the player and experience of the player, which aligns with the literature (Bryson et al., 2013; Carmichael et al., 1999; Carmichael & Thomas, 1993; Dobson & Gerrard, 1997, 1999; Gerrard & Dobson, 2000; Szymanski, 2000; Wicker et al., 2013).

If player cash flows can be accurately measured, the intrinsic valuation process should be the default valuation method. Without player derived cash flows, intrinsic valuations are not possible under this

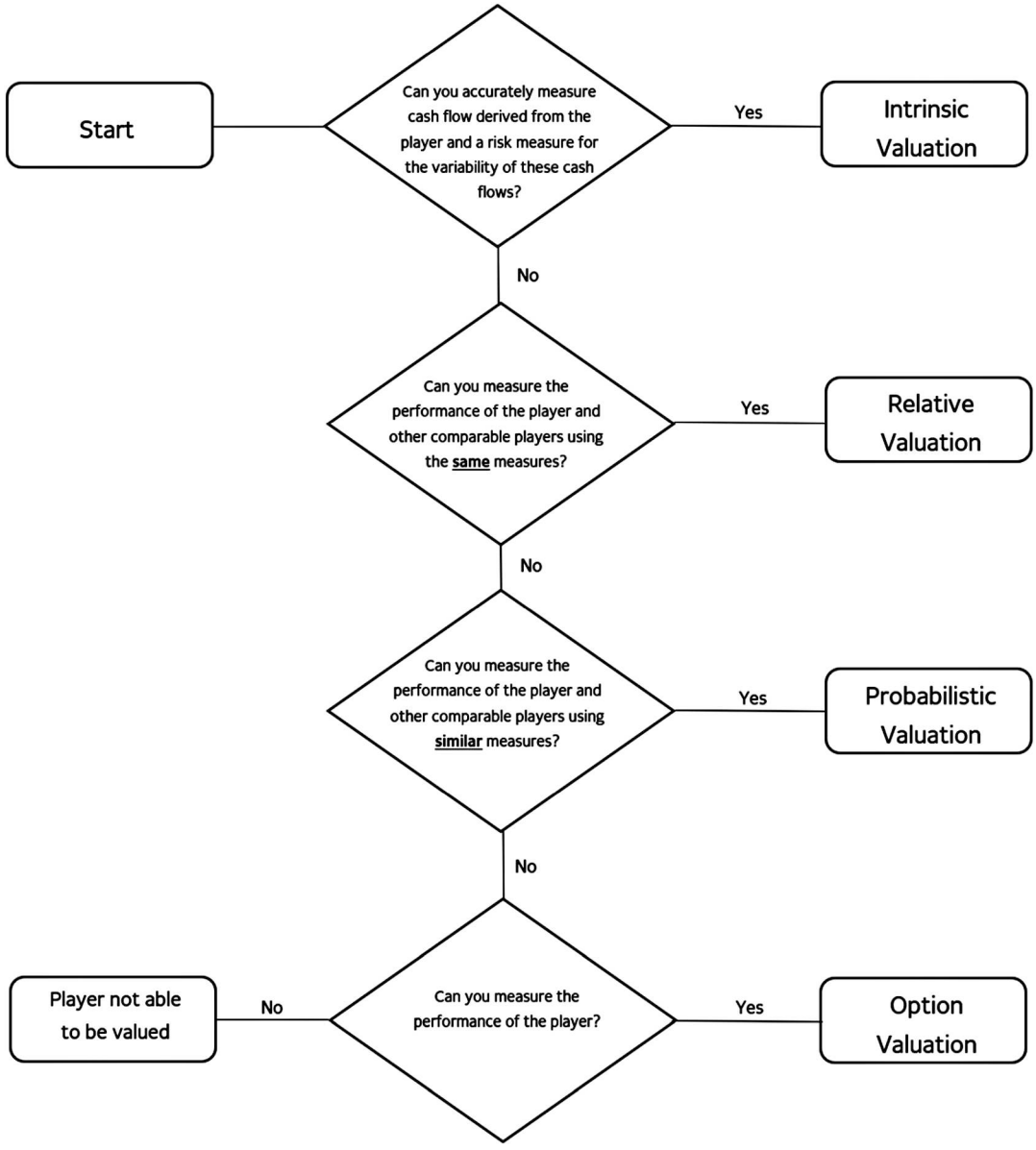


Figure 1. Football player valuation typology decision framework.

framework. Pragmatically, the next valuation process to use would be relative valuation, given that it is a valuation process often used in multiple industries and requires the least amount of technical knowledge. Under Damodaran's (2012, 2018) framework for relative valuation, there were three principles of relative valuation. First, you need to collect a group of comparable assets. Second, estimate measures of standardised variables for this group. Third, controlling for differences between assets in the group and the asset being valued is necessary to ensure the measures are considered reasonable. If the data is not the same across all comparable players, you cannot standardise the measure of the group. Furthermore, without the same variables, it is not possible to control for group differences or develop a

consensus that the players are comparable. Therefore, the decision to use relative valuation depends on using the same variables across all comparable players. Under a relative valuation approach, there is no consideration for cash flow generation. Probabilistic valuation would be the next recommended process if relative valuation is not possible. Modern machine learning methods demonstrate a researcher's ability to develop predictive information from a constrained sample (McHale & Holmes, 2023). If a researcher or practitioner has a sample of data that is believed to be value-driving but not universally available across all players under consideration, then machine learning and other probabilistic techniques could provide a valuation process for football players using that data. For example, a football club will hold more information about players under their employment than one they seek to sign. Should the club develop a performance measure based on the data of their players (e.g. cognitive, leadership, etc.) which appears to correlate with a universal measure of performance (e.g. Goals, Passes, Tackles, etc.), practitioners could develop a valuation proxy based upon that correlated data, seeking to find latent drivers of value. This is not limited to performance data and could apply to known price data (e.g. wages paid to players within the club or transfer fees paid/received) and seek to predict the prices of other players using universal performance measures. In accordance with Damodaran (2012, 2018), consideration for this approach should always be drawn to the likelihood of that player's cash flow generation. The key questions should revolve around how the predicted metric or latent variable adds or subtracts to the predicted cash flow generation of the player. If a researcher or practitioner can assimilate team performance increase with the overall increase in cash flow, this would need to be explicitly stated and demonstrated. Given the difficulties noted with the MRP approach in Scully (1974) and its use in football, this indeed creates a good area for future research. If the club or practitioner is using known prices, then the probabilistic valuation method could predict the values and prices of other players based on this data. This assumes you can accurately control the price driving factors. The decision to move to an option valuation framework would be utilised when researchers or practitioners seek a valuation in an isolated setting, where the market of other players or the information of other players is not dependent on the decision-making process. This approach would be most suitable in negotiation strategies between two parties, performance predictions, future transfer fees, wage predictions or any setting where value is derived from only this player. As with intrinsic and probabilistic valuations and in accordance with Damodaran (2012, p. 2018), option valuation should focus on the cash flow generative aspects of the valuation process.

It should be noted that valuation types do not need to be used in isolation and can be blended, as noted in McHale and Holmes (2023) and Gerrard (2001). For example, an option model can be developed for each player and then used comparatively as long as the variables and method used in the options models are identical. This would not violate the requirements under relative valuation and can support or reinforce assertions derived from the options model. Another example would be a probabilistic model that can support an intrinsic valuation. Providing the measure of cash flow is accurate, probabilistic modelling can be utilised to support risk measure variations of cash flow and therefore varied intrinsic valuations. To support researchers, Table 1 provides definitions of valuation types, potential methods and relevant sources from previous research where the typology and method were used.

In summary, we note the following. With intrinsic valuation approaches in sports, MRP is often utilised. However, we believe the current methodology faces limitations in accurately attributing revenue to a player's contributions to

Table 1. Football player valuation framework.

Valuation Type	Definition	Potential Methods	Literature Examples
Intrinsic	A player who is cash flow generative now and this can be measured against a specific level of risk. Risk in this context would be performance related.	Discounted Cash Flow	Leifheit and Follert (2021)
Relative	A group of comparable players where value measurement variables are standardised and differences are controlled for. Cash flow generation is not relevant.	Comparative, benchmarking.	Berri et al. (2023), Campa (2022), Deutscher and Büschemann (2014), Gerrard (2001), Gerrard (2015), Szymanski and Smith (1997)
Probabilistic	The probability a player could be cash flow generative by projecting or predicting future performance, relative to other players.	Machine Learning, Decision Trees, Game Theory.	Carmichael and Thomas (1993), Pappalardo et al. (2019), Aydemir et al. (2022), McHale and Holmes (2023)
Option	A player who could be cash flow generative in the future but is not cash flow generative now, not relative to other players.	Black-Scholes Options Pricing Model, Markov Process	Tunaru et al. (2005), Tunaru and Viney (2010), Kanyinda et al. (2012) Coluccia et al. (2018)

football. With advancements in computing power and more robust analytical techniques, intrinsic valuation methods can be revisited after addressing key issues: developing a unified metric for player performance, understanding how individual player performance influences team performance, how team performance affects match results and projecting match results to estimate revenues. While theoretically feasible and partially addressed, practical application remains distant.

Relative valuation appears to be the most pragmatic approach for player valuation. By identifying a group of comparable players, standardising variables and controlling for group differences, this method is straightforward to implement. Despite challenges in finding standardised variables and a unified performance metric, the theoretical foundation of this approach is pragmatic. We note that it is crucial to avoid conflating price and valuation or assuming that factors influencing player prices also affect their valuations. While some factors, like performance levels, may overlap, elements such as contract length do not determine value. Similar distinctions apply to buying and selling club characteristics. Researchers must ensure that future studies do not conflate drivers of price with drivers of value and conversely.

Real options present an intriguing avenue for exploration. Despite the extensive literature on options modelling, there is a lack of research in applying this to player valuations. We believe options modelling can be valuable for determining player prices, starting with expected valuations and incorporating contract length, club characteristics and scenario analysis. This approach could offer a robust negotiation framework for clubs, making it a promising area for future price determination and valuation research.

Finally, probabilistic valuations parallel real options but focus on projecting player values and potentially informing price negotiation strategies. With a sufficiently large and accurate data pool, we believe probabilistic methods could enhance contract negotiations for player wages and transfer fees. By utilising a comprehensive performance metric and a training set of player career trajectories, predicting future player performance, values, and prices is possible. This emerging field, as evidenced by the recent studies we outlined that employ machine learning approaches, holds significant potential for future research.

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References

Ackerman, P., & Follert, F. (2019). Some valuation theory comments on the market value analysis of the platform transfermarkt.de. *Sports Und Management*, *3*, 21–41.

Antonioni, P., & Cubbin, J. (2000). The Bosman ruling and the emergence of a single market in soccer talent. *European Journal of Law and Economics*, *9*(2), 157–173. https://doi.org/10.1023/A:1018778718514

Aydemir, A. E., Taskaya Temizel, T., & Temizel, A. (2022). A machine learning ensembling approach to predicting transfer values. *SN Computer Science*, *3*(3), 201. https://doi.org/10.1007/s42979-022-01095-z

Bell, A. R., Brooks, C., & Brooks, R. (2022). Are English football players overvalued? *SSRN Electronic Journal*, *1*, 1–26. https://doi.org/10.2139/SSRN.4080940

Berri, D., Butler, D., Rossi, G., Simmons, R., & Tordoff, C. (2023). Salary determination in professional football: Empirical evidence from goalkeepers. *European Sport Management Quarterly*, *24*(3), 624–640. https://doi.org/10.1080/16184742.2023.2169319

Bikhchandani, S., Hirshleifer, D., & Welch, I. (1992). A theory of fads, fashion, custom, and cultural change as informational cascades. *Journal of Political Economy*, *100*(5), 992–1026. https://doi.org/10.1086/261849

Black, F., & Scholes, M. (1973). The pricing of options and corporate liabilities. *Source: The Journal of Political Economy*, *81*(3), 637–654.

Bodvarsson, ÖB, & Brastow, R. T. (1998). Do employers pay for consistent performance?: Evidence from the NBA. *Economic Inquiry*, *36*(1), 145–160. https://doi.org/10.1111/j.1465-7295.1998.tb01702.x

Bradbury, J. C. (2013). What is right with Scully estimates of a player's marginal revenue product. *Journal of Sports Economics*, *14*(1), 87–96. https://doi.org/10.1177/1527002511418981

Brealey, R. A., Myers, S. C., & Allen, F. (2020). *Principles of corporate finance*. McGraw Hill Education.

Bryson, A., Frick, B., & Simmons, R. (2013). The returns to scarce talent: Footedness and player remuneration in European soccer. *Journal of Sports Economics*, *14*(6), 606–628. https://doi.org/10.1177/1527002511435118

Bryson, A., Rossi, G., & Simmons, R. (2014). The migrant wage premium in professional football: A superstar effect? *Kyklos*, *67*(1), 12–28. https://doi.org/10.1111/kykl.12041

Butler, D., & Coates, D. (2022). Position premium in major league soccer. *International Journal of Sport Finance*, *17*(4), 201–214. https://doi.org/10.32731/ijsf/174.112022.02

Cairns, J., Jennett, N., & Sloane, P. J. (1986). The economics of professional team sports: A survey of theory and evidence. *Journal of Economic Studies*, *13*(1), 3–80. https://doi.org/10.1108/eb002618

Campa, D. (2022). Exploring the market of soccer player registrations: An empirical analysis of the difference between transfer fees and estimated players' inherent value. *Journal of Sports Economics*, *23*(4), 379–406. https://doi.org/10.1177/15270025211059527

Carmichael, F., Forrest, D., & Simmons, R. (1999). The labour market in association football: Who gets transferred and for how much? *Bulletin of Economic Research*, *51*(2), 125–150. https://doi.org/10.1111/1467-8586.00075

Carmichael, F., McHale, I., & Thomas, D. (2011). Maintaining market position: Team performance, revenue and wage expenditure in the english premier league. *Bulletin of Economic Research*, *63*(4), 464–497. https://doi.org/10.1111/j.1467-8586.2009.00340.x

Carmichael, F., & Thomas, D. (1993). Bargaining in the transfer market: Theory and evidence. *Applied Economics*, *25*(12), 1467–1476. https://doi.org/10.1080/00036849300000150

Carrieri, V., Principe, F., & Raitano, M. (2018). What makes you 'super-rich'? New evidence from an analysis of football players' wages. *Oxford Economic Papers*, *70*(4), 950–973. https://doi.org/10.1093/oep/gpy025

Coates, D., Naidenova, I., & Parshakov, P. (2020). Transfer policy and football club performance: Evidence from network analysis. *International Journal of Sport Finance*, *15*(3), 95–109. https://doi.org/10.32731/jsf.2020.a927060

Coates, D., & Parshakov, P. (2022). The wisdom of crowds and transfer market values. *European Journal of Operational Research*, *301*(2), 523–534. https://doi.org/10.1016/j.ejor.2021.10.046

Coluccia, D., Fontana, S., & Solimene, S. (2018). An application of the option-pricing model to the

valuation of a football player in the 'Serie A League'. *International Journal of Sport Management and Marketing*, *18*(1/2), 155–152. https://doi.org/10.1504/IJSMM.2018.091345

Constantinou, A., & Fenton, N. (2017). Towards smart-data: Improving predictive accuracy in long-term football team performance. *Knowledge-Based Systems*, *124*, 93–104. https://doi.org/10.1016/j.knosys.2017.03.005

Damodaran, A. (2006). Valuation approaches and metrics: A survey of the theory and evidence.

Damodaran, A. (2012). *Investment valuation: Tools and techniques for determining the value of any asset. University Edition* (3rd ed., Vol. 1). Wiley.

Damodaran, A. (2018). *The dark side of valuation: Valuing young, distressed and complex buisnesses* (3rd ed., Vol. 1). Pearson FT Press.

Damodaran, A. (2024). Damodaran online. Https://Pages.Stern.Nyu.Edu/~adamodar/

Dawson, P., Dobson, S., & Gerrard, B. (2000). Estimating coaching efficiency in professional team sports: Evidence from English association football. *Scottish Journal of Political Economy*, *47*(4), 399–421. https://doi.org/10.1111/1467-9485.00170

DCMS. (2023). Fact sheet - the Independent Football Regulator (IFR) - GOV.UK. https://www.gov.uk/government/publications/football-governance-bill-supporting-documents/fact-sheet-the-independent-football-regulator-ifr

Depken, C., & Globan, T. (2021). Football transfer fee premiums and Europe's big five. *Southern Economic Journal*, *87*(3), 889–908. https://doi.org/10.1002/soej.12471

Depken, C., & Wilson, D. (2004). Labor markets in the classroom: Marginal revenue product in major league baseball. *Journal of Economics and Finance Education*, *3*(1), 12–24.

Detzen, D., & Löhlein, L. (2023). Valuing soccer players: on the valuation dynamics of an online user community. *Accounting, Auditing & Accountability Journal*, *37*(*2*), 608–626. https://dx.doi.org/10.1108/AAAJ-05-2022-5824

Deutscher, C., & Büschemann, A. (2014). Does performance consistency pay off financially for players? Evidence from the Bundesliga. *Journal of Sports Economics*, *17*(1), 27–43. https://doi.org/10.1177/1527002514521428

Di Domizio, M., Caruso, R., & Frick, B. (2024). The appraisal of players' transfer market values: Empirical evidence from Italian Serie A. *International Journal of Sport Finance*, *19*(1), 39–51. https://doi.org/10.32731/ijsf/191.022024.03

Dobson, S., & Gerrard, B. (1997). *Testing for rent-sharing in soccer transfer fees: Evidence from the English Football League (E97/03)*.

Dobson, S., & Gerrard, B. (1999). The determination of player transfer fees in English professional soccer. *Journal of Sport Management*, *13*(4), 259–279. https://doi.org/10.1123/jsm.13.4.259

El-Hodiri, M., & Quirk, J. (1971). An economic model of a professional sports league. *Journal of Political Economy*, *79*(6), 1302–1319. https://doi.org/10.1086/259837

Fama, E. F. (1970). Efficient capital markets: A review of theory and empirical work. *Journal of Finance*, *25*(2), 383–417. https://doi.org/10.2307/2325486

Fearnhead, P., & Taylor, B. M. (2011). On estimating the ability of NBA players. *Journal of Quantitative Analysis in Sports*, *7*(3), 1–16. https://doi.org/10.2202/1559-0410.1298

Follert, F., & Gleißner, W. (2024). A decision model to value football player investments under uncertainty. *Management Decision*, *62*(13), 178–200. https://doi.org/10.1108/MD-06-2023-0899

Fort, R., Lee, Y. H., & Oh, T. (2019). Quantile insights on market structure and worker salaries: The case of major league baseball. *Journal of Sports Economics*, *20*, 1066–1087. https://doi.org/10.1177/1527002519851152

Fort, R., & Quirk, J. (1995). Cross-subsidization, incentives, and outcomes in professional team sports leagues on JSTOR. *American Economic Association*, *33*(3), 1265–1299. https://www.jstor.org/stable/2729122?seq=1#metadata_info_tab_contents

Franceschi, M., Brocard, J., Follert, F., & Gouguet, J. (2023a). Determinants of football players' valuation: A systematic review. *Journal of Economic Surveys*, *38*(3), 577–600. https://doi.org/10.1111/joes.12552

Franceschi, M., Brocard, J., Follert, F., & Gouguet, J. (2023b). Football players in light of economic value theory: Critical review and conceptualisation. *Managerial and Decision Economics*, *45*(2), 896–920. https://doi.org/10.1002/MDE.4039

Franck, E., & Nüesch, S. (2012). Talent and/or popularity: What does it take to be a superstar? *Economic Inquiry*, *50*(1), 202–216. https://doi.org/10.1111/j.1465-7295.2010.00360.x

Frick, B. (2007). The football players' labor market: Empirical evidence from the major European leagues. *Scottish Journal of Political Economy*, *54*(3), 422–446. https://doi.org/10.1111/j.1467-9485.2007.00423.x

Frick, B. (2011). Performance, salaries and contract length: Empirical evidence from German soccer. *International Journal of Sport Finance*, *6*(2), 87–118.

Frick, B., Barros, C. P., & Prinz, J. (2010). Analysing head coach dismissals in the German "Bundesliga" with a mixed logit approach. *European Journal of Operational Research*, *200*(1), 151–159. https://doi.org/10.1016/j.ejor.2008.11.048

Fumarco, L., & Rossi, G. (2018). The relative age effect on labour market outcomes – Evidence from Italian football. *European Sport Management Quarterly*, *18*(4), 501–516. https://doi.org/10.1080/16184742.2018.1424225

Galton, F. (1907). Vox Populi. *Nature*, *75*(1949), 450–451. https://doi.org/10.1038/075450a0

Gerrard, B. (2001). A new approach to measuring player and team quality in professional team sports. *European Sport Management Quarterly*, *1*(3), 219–234. https://doi.org/10.1080/16184740108721898

Gerrard, B. (2007). Is the moneyball approach transferable to complex invasion team sports?: EBSCOhost. *International Journal of Sport Finance*, *2*, 214–230. https://web.s.ebscohost.com/ehost/pdfviewer/pdfviewer?vid=1&sid=16424833-b1ac-49db-9b58-7e628d7f79b4%40redis

Gerrard, B. (2015). Rigour and relevance in sport management: Reconciling the competing demands of disciplinary research and user-value. *European Sport Management Quarterly*, *15*(5), 505–515. https://doi.org/10.1080/16184742.2015.1085714

Gerrard, B., & Dobson, S. (2000). Testing for monopoly rents in the market for playing talent – Evidence from English professional football. *Journal of Economic Studies*, *27*(3), 142–164. https://doi.org/10.1108/01443580010326049

Haug, E. G., & Taleb, N. N. (2011). Option traders use (very) sophisticated heuristics, never the Black-Scholes-Merton formula. *Journal of Economic Behavior & Organization*, *77*(2), 97–106. https://doi.org/10.1016/j.jebo.2010.09.013

He, M., Cachucho, R., & Knobbe, A. (2015). *Football player's performance and market value*. Proceedings of the 2nd Workshop of Sports Analytics, European Conference on Machine Learning and Principles and Practice of Knowledge Discovery in Databases (ECML PKDD).

Herm, S., Callsen-Bracker, H. M., & Kreis, H. (2014). When the crowd evaluates soccer players' market values: Accuracy and evaluation attributes of an online community. *Sport Management Review*, *17*(4), 484–492. https://doi.org/10.1016/j.smr.2013.12.006

Jones, J. C. H. (1969). The economics of the National Hockey League. *Canadian Economic Association*, *2*(1), 1–20. https://doi.org/10.2307/133568

Kadane, J. B., & Larkey, P. D. (1982). Subjective probability and the theory of games. *Management Science*, *28*(2), 113–120. https://doi.org/10.1287/mnsc.28.2.113

Kanyinda, A., Bouteiller, C., & Karyotis, C. (2012). Human capital: Assessing the financial value of football players on the basis of real options theory. *Investment Management and Financial Innovations*, *9*(4), 27–37.

Kesenne, S. (2000). Revenue sharing and competitive balance in professional team sports. *Journal of Sports Economics*, *1*(1), 56–65. https://doi.org/10.1177/152700250000100105

Kharrat, T., McHale, I. G., & Peña, J. L. (2020). Plus–minus player ratings for soccer. *European Journal of Operational Research*, *283*(2), 726–736. https://doi.org/10.1016/j.ejor.2019.11.026

Koller, T., Dobbs, R., & Huyett, B. (2010). *Value: The four cornerstones of corporate finance (1st ed., Vol. 1)*. Wiley Publishing.

Koller, T., Goedhart, M. H., & Wessels, D. (2020). *Valuation: Measuring and managing the value of companies* (7th ed., Vol. 1). Wiley Publishing.

Krautmann, A. C. (2013). What is right with Scully estimates of a player's marginal revenue product: Reply. *Journal of Sports Economics*, *14*(1), 97–105. https://doi.org/10.1177/1527002511419299

Krautmann, A. C. (2018). Contract extensions: The case of major league baseball. *Journal of Sports Economics*, *19*(3), 299–314. https://doi.org/10.1177/1527002516656727

Leeds, M. A., & Kowalewski, S. (2001). Winner take all in the NFL: The effect of the salary cap and free agency on the compensation of skill position players. *Journal of Sports Economics*, *2*(3), 244–256. https://doi.org/10.1177/152700250100200304

Leifheit, N., & Follert, F. (2021). Financial player valuation from the perspective of the club: The case of football. *Managing Sport and Leisure*, *28*(3), 618–637. https://doi.org/10.1080/23750472.2021.1944821

Lucifora, C., & Simmons, R. (2003). Superstar effects in sport: Evidence from Italian soccer. *Journal of Sports Economics*, *4*(1), 35–55. https://doi.org/10.1177/1527002502239657

Macdonald, B. (2012). Adjusted plus-minus for NHL players using ridge regression with goals, Shots, Fenwick, and Corsi. *Journal of Quantitative Analysis in Sports*, *8*(3), 1–24. https://doi.org/10.1515/1559-0410.1447

Martín-Magdalena, J., De los Ríos-Sastre, S., Redondo, R., & Alaminos, D. (2024). Effectiveness of UEFA's regulation for European football financial management: A comprehensive systematic review and meta-analysis. *Heliyon*, *10*(20), e39151. https://doi.org/10.1016/j.heliyon.2024.e39151

McHale, I. G., & Holmes, B. (2023). Estimating transfer fees of professional footballers using advanced performance metrics and machine learning. *European Journal of Operational Research*, *306*(1), 389–399. https://doi.org/10.1016/j.ejor.2022.06.033

Medcalfe, S., & Smith, R. (2018). The effect of foreign players on pay and performance in major league soccer. *International Journal of Sport Finance*, *13*(4), 297–318.

Merton, R. C. (1973). Theory of rational option pricing. *Source: The Bell Journal of Economics and Management Science*, *4*(1), 141–183. https://www.jstor.org/stable/3003143

Metelski, A. (2021). Factors affecting the value of football players in the transfer market. *Journal of Physical Education and Sport*, *21*, 1150–1155. https://doi.org/10.7752/jpes.2021.s2145

Monteiro, R. K., Prates, R. C., & Frota, L. M. (2023). The determinants of player transfers in Brazil: The role of expectations in the football market. *Applied Economics*, *55*(26), 2964–2977. https://doi.org/10.1080/00036846.2022.2107989

Müller, O., Simons, A., & Weinmann, M. (2017). Beyond crowd judgments: Data-driven estimation of market value in association football. *European Journal of Operational Research*, *263*(2), 611–624. https://doi.org/10.1016/j.ejor.2017.05.005

Myers, S. C. (1977). Determinants of corporate borrowing. *Journal of Financial Economics*, *5*(2), 147–175. https://doi.org/10.1016/0304-405X(77)90015-0

Nash, J. (1950). The bargaining problem. *Econometrica*, *18*(*2*), 155–162. https://doi.org/10.2307/1907266

Nash, J. (1953). Two-person cooperative games. *Econometrica*, *21*(*1*), 128–140. https://doi.org/10.2307/1906951

Neale, W. C. (1964). The peculiar economics of professional sports: A contribution to the theory of the firm in sporting competition and in market competition. *The Quarterly Journal of Economics*, *78*(1), 1–14. https://doi.org/10.2307/1880543

Pappalardo, L., Cintia, P., Ferragina, P., Massucco, E., Pedreschi, D., & Giannotti, F. (2019). PlayeRank: Data-driven performance evaluation and player ranking in soccer via a machine learning approach. *ACM Transactions on Intelligent Systems and Technology*, *10*(5), 1–27. https://doi.org/10.1145/3343172

Piovaccari, G. (2023). *Juventus to pay $790,000 fine, no further points deducted*. Reuters.

Plumley, D., Ramchandani, G. M., & Wilson, R. (2019). The unintended consequence of Financial Fair Play. *Sport, Business and Management: An International Journal*, *9*(2), 118–133. https://doi.org/10.1108/SBM-03-2018-0025

Poli, R., Besson, R., & Ravenel, L. (2022). Econometric approach to assessing the transfer fees and values of professional football players. *Economies*, *10*(1), 4. https://doi.org/10.3390/economies10010004

Premier League. (2023). *History and records*. https://www.premierleague.com/history/dashboard

Prockl, F., & Frick, B. (2018). Information precision in online communities: Player valuations on www.transfermarkt.de. *International Journal of Sports Finance*, *13*(4), 319–335.

Quansah, T., Frick, B., Lang, M., & Maguire, K. (2021). The importance of club revenues for player salaries and transfer expenses—How does the coronavirus outbreak (COVID-19) impact the english premier league? *Sustainability*, *13*(9), 5154. https://doi.org/10.3390/su13095154

Richardson, D. H. (2000). Pay, performance and competitive balance in the National Hockey League. *Eastern Economic Journal*, *26*(4), 393–417. https://web.holycross.edu/RePEc/eej/Archive/Volume26/V26N4P393_417.pdf

Richau, L., Follert, F., Frenger, M., & Emrich, E. (2019). Performance indicators in football: The importance of actual performance for the market value of football players. *Sporta Saintika*, *4*(1), 41–67. https://doi.org/10.24036/sporta.v4i1.98

Rodrigues, F., & Pinto, Â. (2022). Prediction of football match results with Machine Learning. *Procedia Computer Science*, *204*, 463–470. https://doi.org/10.1016/j.procs.2022.08.057

Rottenberg, S. (1956). The baseball players' labor market. *Source: Journal of Political Economy*, *64*(3), 242–258. https://about.jstor.org/terms

Rubio Martín, G., Manuel García, C. M., Rodríguez-López, Á, & Gonzalez Sanchez, F. J. (2022). Measuring football clubs' human capital: Analytical and dynamic models based on footballers' life cycles. *Journal of Intellectual Capital*, *23*(5), 1107–1137. https://doi.org/10.1108/JIC-06-2020-0211

Saebø, O. D., & Hvattum, L. M. (2019). Modelling the financial contribution of soccer players to their

clubs. *Journal of Sports Analytics*, *5*(1), 23–34. https://doi.org/10.3233/JSA-170235

Samuelson, P. A. (1965). Rational theory of warrant pricing. *Industrial Management Review*, *6*(2), 13–39.

Sanderson, A. R., & Siegfried, J. J. (2006). Simon Rottenberg and baseball, then and now: A fiftieth anniversary retrospective. *Journal of Political Economy*, *114*(3), 594–605. https://doi.org/10.1086/503756

Scott, F. A., Long, J. E., & Somppi, K. (1985). Salary Vs. marginal revenue product under monopsony and competition: The case of professional basketball. *Atlantic Economic Journal*, *13*(3), 50–59. https://doi.org/10.1007/BF02304294

Scully, G. W. (1974). Pay and performance in major league baseball. *American Economic Review*, *64*(6), 915–930.

Serna Rodríguez, M., Ortega Alvarez, A. M., & Arango-Vasquez, L. (2022). Worldwide trends in the scientific production on soccer players market value, a bibliometric analysis using bibliometrix R-tool. *Team Performance Management: An International Journal*, *28*(5/6), 415–440. https://doi.org/10.1108/TPM-02-2022-0015

Serna Rodríguez, M., Ramírez Hassan, A., & Coad, A. (2019). Uncovering value drivers of high performance soccer players. *Journal of Sports Economics*, *20*(6), 819–849. https://doi.org/10.1177/1527002518808344

Simmons, R. (1997). Implications of the Bosman ruling for football transfer markets. *Economic Affairs*, *17*(3), 13–18. https://doi.org/10.1111/1468-0270.00036

Simmons, R. (2021). Professional labor markets in the journal of sports economics. *Journal of Sports Economics*, *23*, 728–748. https://doi.org/10.1177/15270025211051062

Simmons, R., & Berri, D. J. (2011). Mixing the princes and the paupers: Pay and performance in the National Basketball Association. *Labour Economics*, *18*(3), 381–388. https://doi.org/10.1016/j.labeco.2010.11.012

Singh, P., & Lamba, P. S. (2019). Influence of crowdsourcing, popularity and previous year statistics in market value estimation of football players. *Journal of Discrete Mathematical Sciences and Cryptography*, *22*(2), 113–126. https://doi.org/10.1080/09720529.2019.1576333

Sloane, P. J. (1969). The labour market in professional football*. *British Journal of Industrial Relations*, *7*(2), 181–199. https://doi.org/10.1111/j.1467-8543.1969.tb00560.x

Sloane, P. J. (1971). The economics of professional football: The football club as a utility maximiser. *Scottish Journal of Political Economy*, *18*(2), 121–146. https://doi.org/10.1111/j.1467-9485.1971.tb00979.x

Sloane, P. J. (2006). Rottenberg and the economics of sport after 50 years: An evaluation. *SSRN Electronic Journal*, 2–23. https://doi.org/10.2139/ssrn.918719

Solow, J. L., & Krautmann, A. C. (2020). Do you get what you pay for? Salary and ex ante player value in major league baseball. *Journal of Sports Economics*, *21*(7), 705–722. https://doi.org/10.1177/1527002520930259

Surowiecki, J. (2004). *The wisdom of crowds : Why the many are smarter than the few and how collective wisdom shapes business, economies, societies, and nations*. Doubleday.

Szymanski, S. (2000). A market test for discrimination in the English professional soccer leagues. *Journal of Political Economy*, *108*(3), 590–603. https://doi.org/10.1086/262130

Szymanski, S. (2001). Income inequality, competitive balance and the attractiveness of team sports: Some evidence and a natural experiment from English soccer. *The Economic Journal*, *111*(469), F69–F84. https://doi.org/10.1111/1468-0297.00599

Szymanski, S., & Smith, R. (1997). The English football industry: Profit, performance and industrial structure. *International Review of Applied Economics*, *11*(1), 135–153. https://doi.org/10.1080/02692179700000008

Thrane, C. (2019). Performance and actual pay in Norwegian soccer. *Journal of Sports Economics*, *20*(8), 1051–1065. https://doi.org/10.1177/1527002519851146

Tunaru, R., Clark, E., & Viney, H. (2005). An option pricing framework for valuation of football players. *Review of Financial Economics*, *14*(3–4), 281–295. https://doi.org/10.1016/j.rfe.2004.11.002

Tunaru, R., & Viney, H. (2010). Valuations of soccer players from statistical performance data. *Journal of Quantitative Analysis in Sports*, *6*(2), 1–21. https://doi.org/10.2202/1559-0410.1238

Tversky, A., & Kahneman, D. (1974). Judgment under uncertainty: Heuristics and biases. *New Series*, *185*(4157), 1124–1131.

Vrooman, J. (2000). The economics of American sports leagues. *Scottish Journal of Political Economy*, *47*(4), 364–398. https://doi.org/10.1111/1467-9485.00169

Weatherall, J. O. (2018). The peculiar logic of the black-scholes model. *Philosophy of Science*, *85*(5), 1152–1163. https://doi.org/10.1086/699677

Wicker, P., Prinz, J., Weimar, D., Deutscher, C., & Upmann, T. (2013). No pain, no gain? Effort and productivity in professional soccer. *International Journal of Sport Finance*, *8*(2), 124–139.

Yang, Y., Koenigstorfer, J., & Pawlowski, T. (2024). Predicting transfer fees in professional European football before and during COVID-19 using machine learning. *European Sport Management Quarterly*, *24*(3), 603–623. https://doi.org/10.1080/16184742.2022.2153898

Yiğit, A. T., Samak, B., & Kaya, T. (2020). Football player value assessment using machine learning techniques. *Advances in Intelligent Systems and Computing*, *1029*, 289–297. https://doi.org/10.1007/978-3-030-23756-1_36